STATEMENT OF WORK

Cryogen-free Dilution Refrigerator Testbed for Characterization of Superconducting Detectors and Devices, Detector Systems Branch, NASA GSFC Code 553

INTRODUCTION:

This Statement of Work (SOW) describes our intent to purchase the standard dilution refrigerator testbed described in the main section of this SOW, with the option to purchase any of the additional options listed in the 'Requirements' section within FY26.

The Detector Systems Branch of NASA Goddard Space Flight Center requires a rapid cycle cryogen free dilution refrigerator (DR) testbed capable of achieving ultralow temperatures and doing so with a high cooling power. This project is undertaken in support of NASA's mission to develop low noise cryogenic detectors and detector system components. Performance characterization and calibration of the DC, Radio-Frequency (RF) and optical response of these detectors and readout components is necessary for their eventual implementation in science missions. Specifically, this project has been undertaken to provide a critically necessary testbed for characterization of low noise superconducting detector components being developed for quantum sensing applications.

Ideally, this test platform shall be designed to give access to the 0.01 Kelvin to 3 Kelvin temperature range within 24-48 hours of mounting devices in the system. The system in its testing configuration shall achieve a base temperature below 10 milliKelvin (mK) with no heat load (except testbed wiring and thermometry) and shall maintain continuous temperatures below 100 mK with a 250 microwatt (uW) heatload. The system shall provide both sufficient cooling power and a suitable electromagnetic and radiation leak-tight environment. The system shall operate without use of liquid cryogens for cooling or precooling of the system or its components. The system shall be built for robustness, with the expectation that it shall perform within specifications for temperature and temperature control over an extended period with limited or no maintenance. The system shall be designed with a rapid duty cycle (warm for sample loading to cryogenic temperatures and back to warm) such that a complete cooldown and warmup can be conducted in a calendar week to support the rate of work anticipated for the system once installed at GSFC.

OBJECTIVES:

Under this agreement, NASA GSFC will obtain a refrigeration system that cools to a <10 mK base temperature and with >250 microWatt cooling power near 100 mK and >10 microWatt cooling power at 20mK (with minimal heatload) without the use of liquid cryogens (such as liquid nitrogen and liquid helium, except for the use of liquid nitrogen in cold traps).

The system performance shall be demonstrated by the vendor at NASA GSFC prior to acceptance by the GSFC technical representative.

REQUIREMENTS:

This SOW defines the requirements for the laboratory layout and design, fabrication, delivery, installation, and commissioning of a cryogen-free high-performance dilution refrigerator system with <10mK base temperature and enhanced vibration isolation features as detailed in the requirements. The vendor support shall include, but is not limited to, the following:

- Delivery of a complete dilution refrigerator system including:
  - Cryostat with installed dilution unit
  - Pulse tube cooler with remote motor capable of achieving <4K base temperature during operation configured to operate with 220V/3PH/60Hz
  - Gas handling system to circulate Helium-3/Helium-4 mixture with features to limit vibration transferred to both the cryostat and the surrounding floor and electrically isolated from cryostat and with sufficient safeguards to prevent the accidental venting of the dilution mixture
  - Control electronics and software for the monitoring of all cryostat parameters including mixture flow rates, system pressures and temperatures and the statuses of various peripheral equipment
  - Thermometry system with a 6 or more channel readout capable of low noise operation for Ruthenium Oxide or similar thermometers
  - All necessary pumps, valves, and peripherals configured to operate with 120V/60Hz
  - Cryostat stand with passive vibration isolation and means for fixturing to floor of laboratory
  - GSFC will supply the requisite Helium-3 needed to achieve normal operation of the system up to 25L STP.

- Technical Requirements:
  - Cryostat shall achieve a base temperature of <10mK
  - Cryostat shall provide > 10uW of cooling power at 20mK and >250uW of cooling power at 100mK
  - Cryostat shall have an experimental volume surrounding the mixing chamber of at least 250mm D x 250mm H
  - Cryostat hould be able to cool from room temperature to base within 48 hours
  - Cryostat shall contain a pulse tube cryocooler capable of achieving a <4K base temperature during operation
  - Pulse tube cryocooler shall be electrically (dielectric unions) and mechanically isolated from cryostat (bellows, flexible heat straps, etc).
  - Pulse tube cryocooler shall have a remote mounted motorized valve
  - Pulse tube cryocooler shall have a over-pressure relief valve as necessary to prevent over-pressure situations.

- Cryostat shall be equipped with radiation shields at ~50K, ~4K, and at the Still stages, with the Still radiation shield being gold plated.
- Cryostat radiation shield mounting surfaces and plates shall be designed to prevent light leaks and have an intentionally designed path for vacuum evacuation.
- Cryostat shall be equipped with vacuum shells with bayonets to allow for single person assembly and disassembly of the cryostat.
- Above mentioned radiation shields (50K, 4K, Still) and cryostat vacuum shell shall have a removeable flat bottom to allow for future modification
- Cryostat shall be constructed in such a manner as to limit the use of magnetic materials near the experimental volume, with the materials inside and adjacent to the experimental volume being non-magnetic (e.g. Copper, Brass, 316 Stainless Steel, Non-magnetic gold platings, etc).
- Cryostat shall be equipped with a dilution unit that is designed to limit to an absolute minimum the use of soft soldered or indium seals to avoid superfluid He leaks and require more than 25L of STP Helium-3 to achieve normal operation.
- System shall be delivered with a gas handling sub-system that ensures the continuous flow of the Helium-3/Helium-4 gas mixture for the condensation and operation of the dilution fridge.
- The gas handling system shall be electrically and mechanically isolated from the cryostat, as well as constructed in a manner to prevent vibrations from being transferred to the floor.
- System shall be equipped with a liquid nitrogen cold trap or internal cold trap to prevent contamination in the dilution unit as necessary.
- Cryostat shall contain integrated thermometry at the ~50K, ~4K, Still and Mixing Chamber stages with temperature appropriate thermometers at each stage.
- Cryostat shall contain heat switches as necessary to facilitate the cooldown of the Still and Mixing Chamber thermal masses
- Cryostat shall contain heaters as necessary to operate the dilution unit, as well as a heater at the mixing chamber to facilitate temperature control, and a heater at the ~4K stage to allow faster warm-up.
- Cryostat system shall include a thermometry readout of at least 6 channels (4 listed above + 2 extra) with electronics appropriate for the measurement of milliKelvin resistances as needed. Thermometry shall also allow for the PID control of the mixing chamber stage with temperature stability of 0.1mK RMS or better. The thermometry system shall be able to be interfaced using the Python programming language, either directly or through a robust web interface.
- Cryostat system shall include control electronics and software to operate the gas-handling system and cryostat components.
- Cryostat control system shall allow for remote control the cryostat including the evacuation, initial cooldown, condensation, mixture recapture and warm-up of the system. Automation scripts for these processes shall be included.

- Cryostat shall be equipped with at least 2 large bore (>60mm ID) line of sight access ports to allow for the installation of electronics and wiring in the cryostat.
- Cryostat shall be equipped with 2x 12 Twisted Pair Cables that pass through from a connectorized bulkhead at the vacuum shell, are sufficiently heat sunk at each cryogenic stage, and terminate in micro-Dsub 25 (MDM25) connectors at the mixing chamber stage. The cables shall be shielded with a stainless steel fine braid and have a MDM25 connectorized break at the ~4K stage. From room temperature to ~4K, the cables shall be constructed from Phsophor Bronze twisted pair wire, while from ~4K to the mixing chamber, the cables shall be constructed from NbTi superconducting alloy twisted pair wire.
- Cryostat shall be mounted in a stand designed to passively damp vibrations arising from the pulse tube cryocooler an remote motor valve. The resultant vibrational level at the mixing chamber shall be <10um/rtHz for frequencies <20Hz, <1um/rtHz for frequencies >20Hz and <100Hz, and <0.1um/rtHz for frequencies >100Hz. If an active vibration isolation system is needed to meet these specifications, the system shall be equipped as such.
- The system shall be equipped to shutdown and restart safely after brief power outages without permanent loss of any of the Helium-3 charge.

- Design guidance for the installation of the system at GSFC
- On-site installation and commissioning of the system.
- The vendor shall be equipped to provide on-going technical support as needed to diagnose issues with the system for the life of the system.
- The vendor shall provide at minimum a 1-year warranty for parts and service of cryogen free refrigeration system.

ACCEPTANCE CRITERIA:

- The dilution refrigerator system shall be deemed acceptable when system is constructed and installed at GSFC including all of the following:
    - Successful He leak checking of vacuum system and dilution unit
    - Base temperature <10 mK is demonstrated at the mixing chamber
    - Cooling power of at least 250 µW at 100 mK is verified
    - Nominal temperatures of the 50K, 4K and Still during operation
    - If active vibration isolation is present, vibration isolation must meet the specified requirements
    - All documentation is delivered including cryostat test performance reports, calibration files, maintenance procedures, drawings and computer aided design (CAD) files
    - Basic user training is completed
    - System operates continuously for 24 hours without faults
    - Any abnormal system operation – vendor will provide the necessary services (cost covered by vendor) until normal operation is obtained.

DELIVERABLES/DELIVERY SCHEDULE:

System is to be installed and commissioned by vendor to verify performance as installed. The deliverables for this work are:

- one (1) cryogen-free dilution refrigeration system, and supporting electrical and gas handling components to enable refrigeration (compressor for pulse tube cooler, He-3/He-4 gas handling system)
- supporting electronics for readout and control of thermometry, heaters and other system components
- additional experimental DC wiring, RF cables and magnetic shielding installed in the system if any of these options are selected
- hardware for mechanical support of the system and its components

Item	Description	Schedule	Delivery Method/Addresses
1	Vendor presents design for approval	Two (2) weeks from date of award	GSFC Technical Representative, Dr. Jake Connors, Code 553.0
2	Vendor informs of on-site requirements for installation at GSFC	One (1) month from date of award	GSFC Technical Representative, Dr. Jake Connors, Code 553.0
3	Vendor gives update on construction progress and notice of any adjustment to estimated delivery timescale	Six (6) months from date of award and/or as needed	GSFC Technical Representative, Dr. Jake Connors, Code 553.0
4	Vendor completes construction and delivery of system	Eleven (11) months from date of award	GSFC Technical Representative, Dr. Jake Connors, Code 553.0
5	Vendor demonstrates system performance at GSFC for final acceptance of system	Twelve (12) months from date of award	GSFC Technical Representative, Dr. Jake Connors, Code 553.0

GOVERNMENT-FURNISHED EQUIPMENT AND GOVERNMENT-FURNISHED INFORMATION:

For testing and acceptance at GSFC, lab space and supporting equipment (pumping station, leak checker, electrical power, cooling water, etc.) will be provided in accordance with requirements provided by the manufacturer.

He-3 will be shipped to vendor location or supplied upon delivery of DR at GSFC for vendor installation into the DR in accordance with requirements provided by the manufacturer.

PERIOD OF PERFORMANCE:

Anticipated period of performance for completion of all work is 12 months from date of award plus an additional 12 months for warranty coverage.